Recursive State Estimation for Dynamic Systems

SoSe2023

Comparison of Several State-of-the-Art filtering methods for position estimation of a moving vehicle

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TASKS

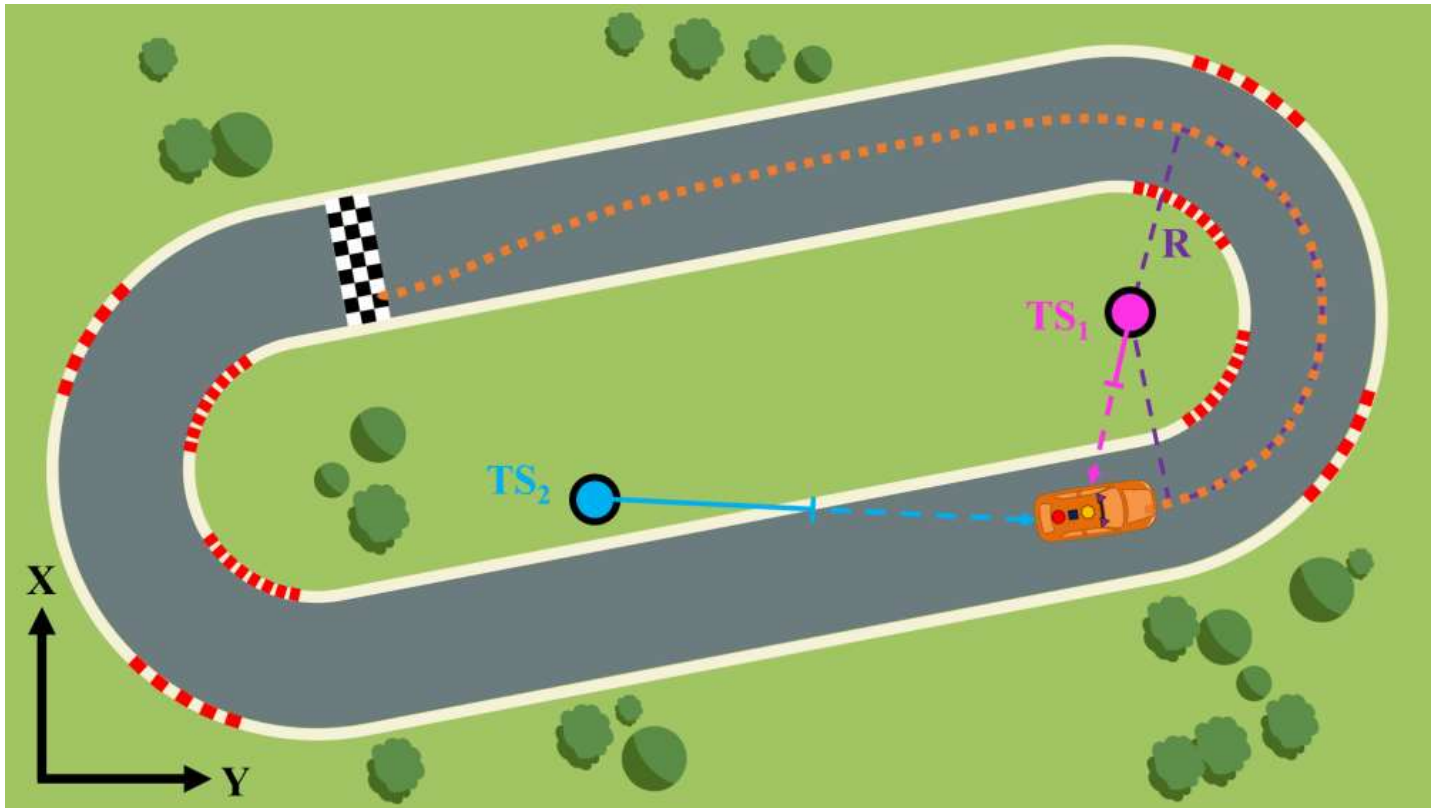

Figure 1

Major Algorithm Step

Calculate :

State Vector, X

Variance Co-variance Matrix (VCM), Q_{xx}

- Intialization

$$\hat{x}_{+}^{(0)}, \hat{Q}_{xx+}^{(0)}$$

- a-priori estimation (prediction step)

$$\hat{x}_{-}^{(k)}, \hat{Q}_{xx-}^{(k)}$$

- a-posterri estimation (update step)

$$\hat{x}_{+}^{(k)}, \hat{Q}_{xx+}^{(k)}$$

Filters	Optimization
Extended Kalman Filter (EKF) Extended Kalman Filter Constraint (EKFC) Iterative Extended Kalman Filter (IEKF) Iterative Extended Kalman Filter Constraint (IEKFC)	First Order Derivative
Unscented Kalman Filter (UKF)	Not Randomly Generated (Deterministically)
Ensembled Kalman Filter (EnKF)	Randomly Generated (Stochastically)
Particle Filter(PF)	Monte Carlo Simulation

Trajectory of the vehicle

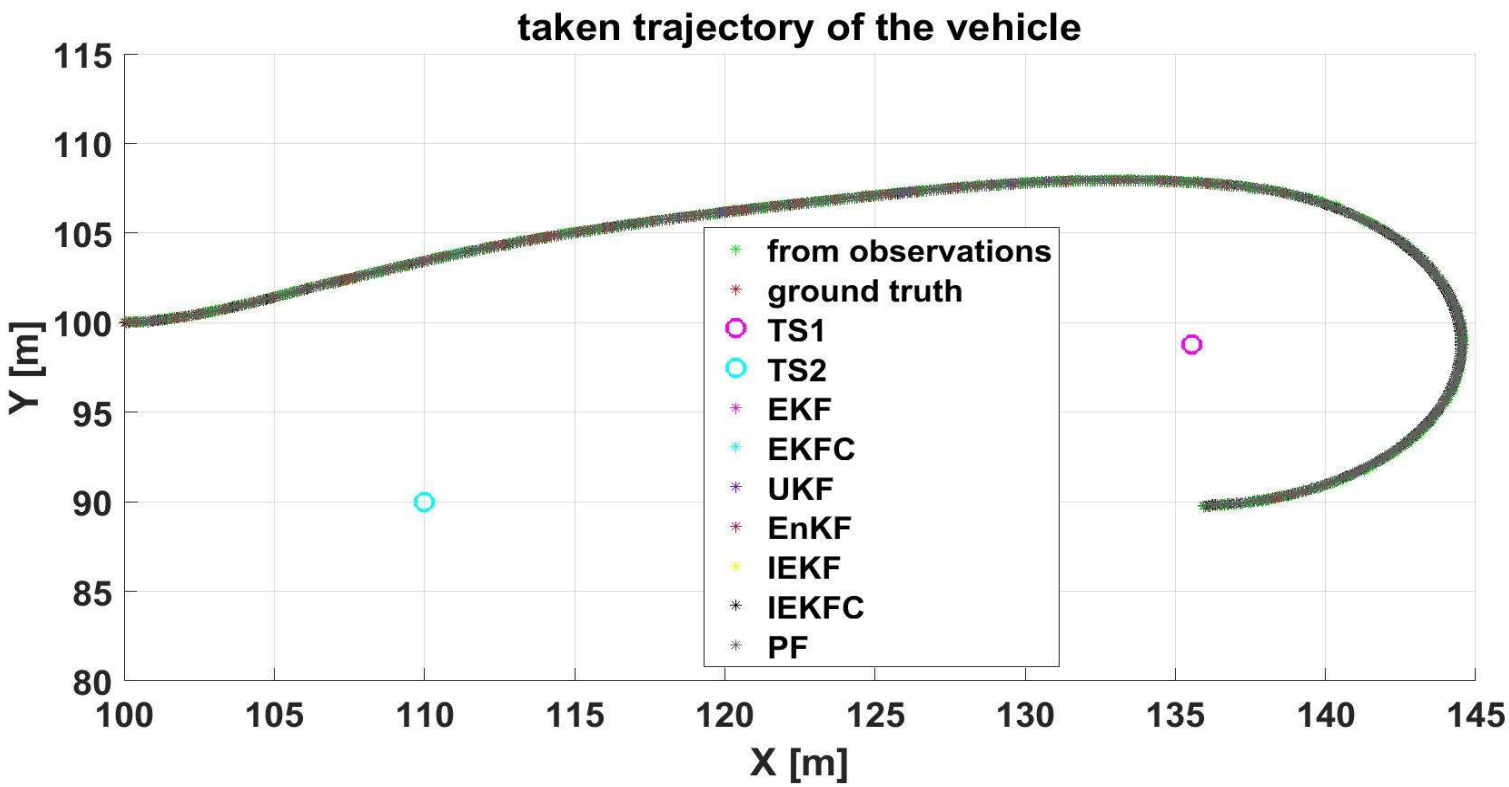

Figure 2

Root Mean Square Error for different Filtering approaches

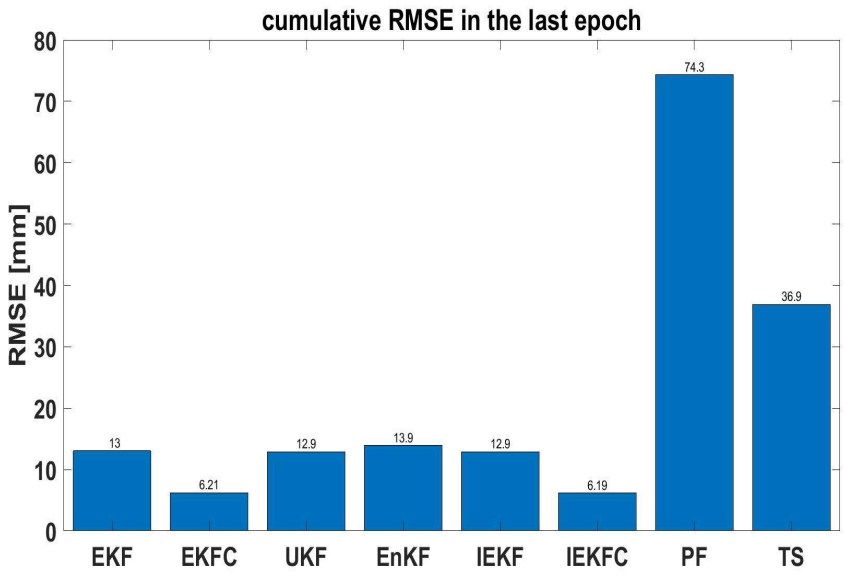

Figure 3

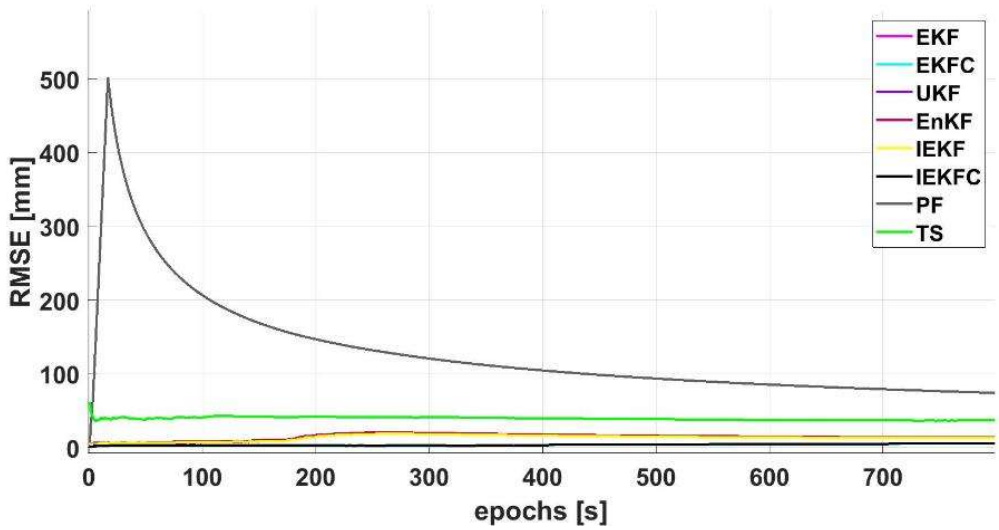

Figure 4

Iterative Extended Kalman Filter(IEKF)

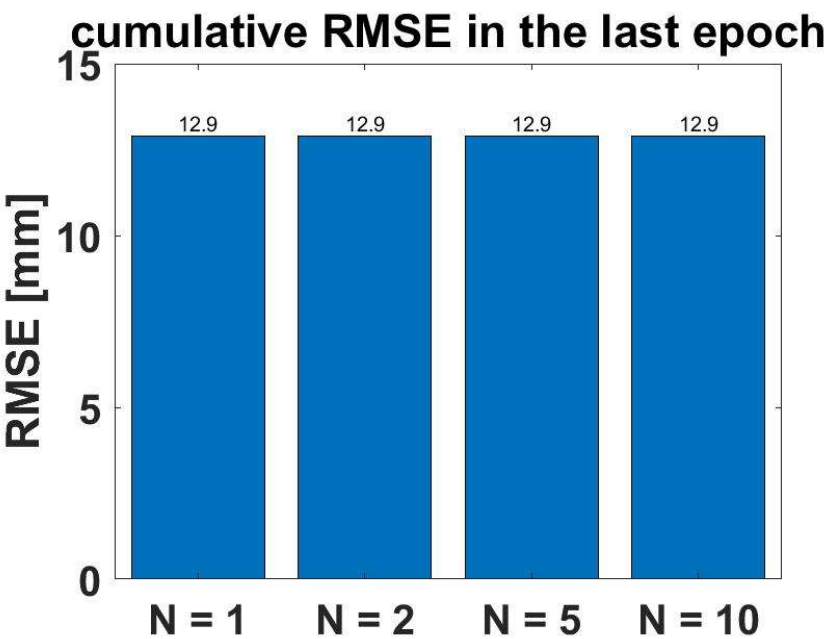

Figure 5

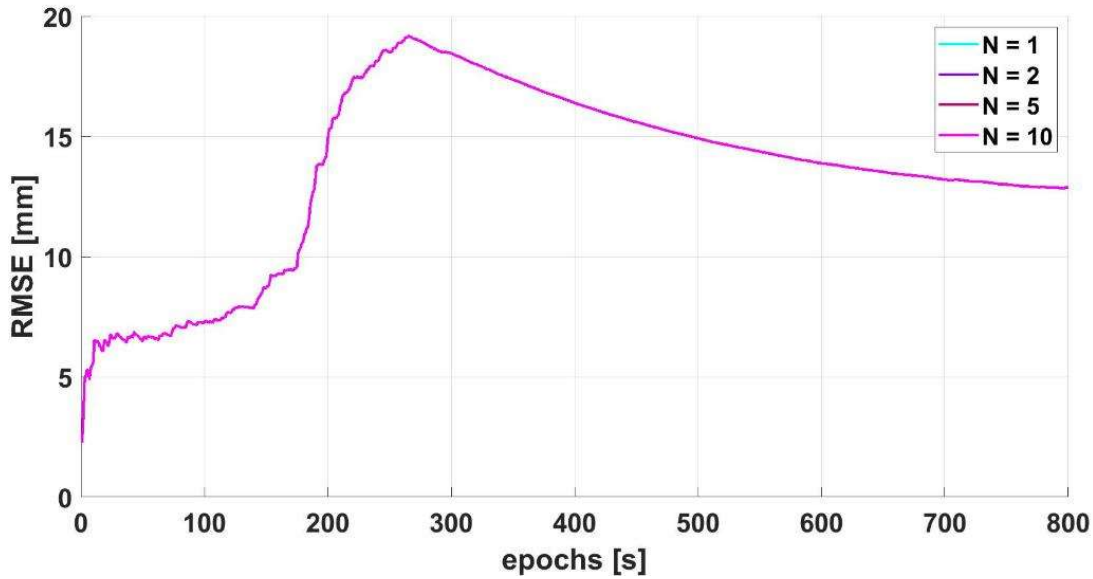

Figure 6

Unscented Kalman Filter(UKF)

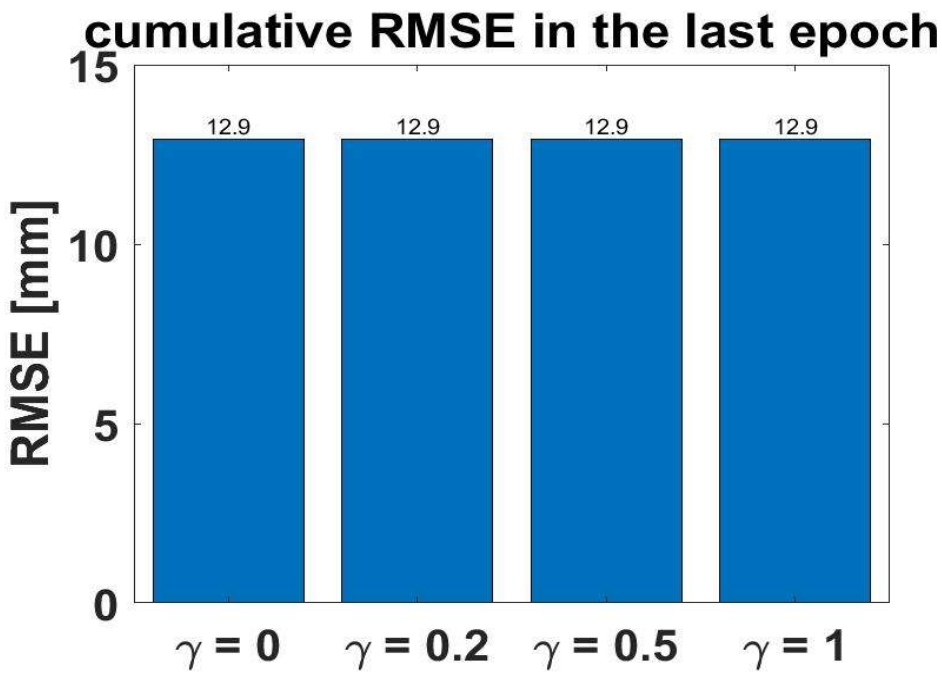

Figure 7

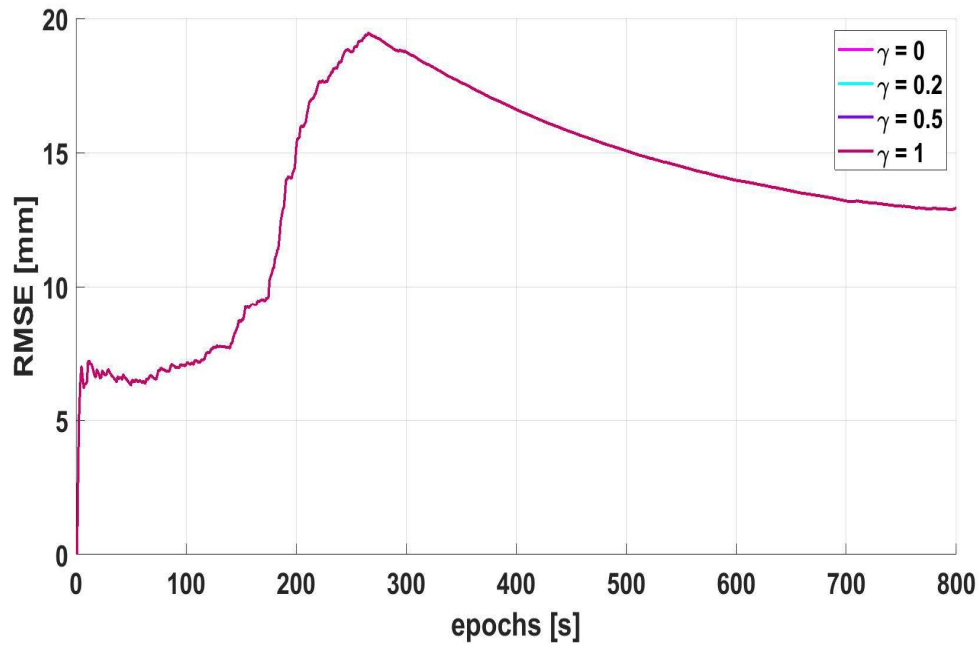

Figure 8

Ensemble Kalman Filter(EnKF)

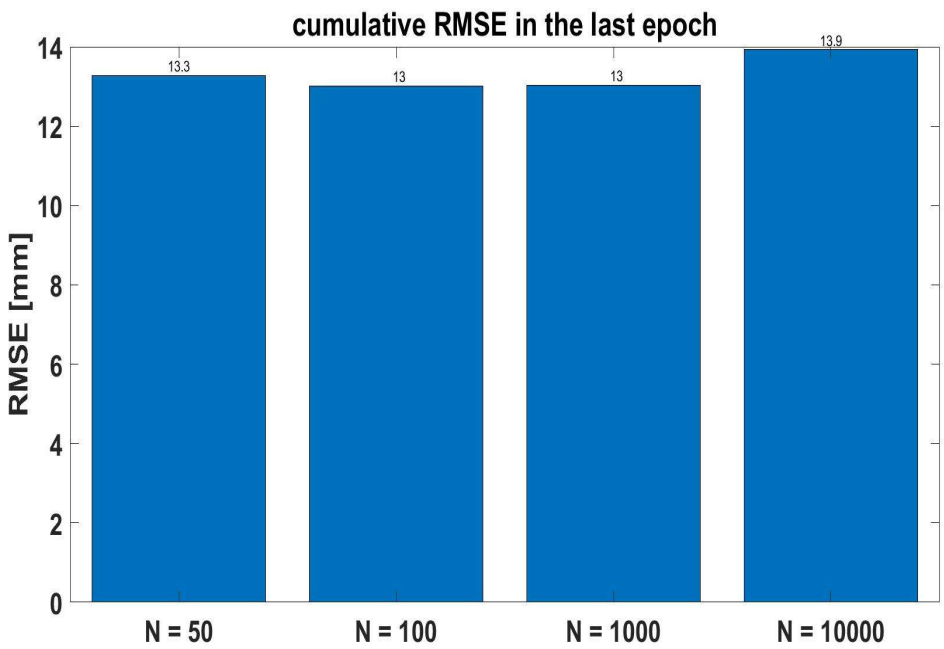

Figure 9

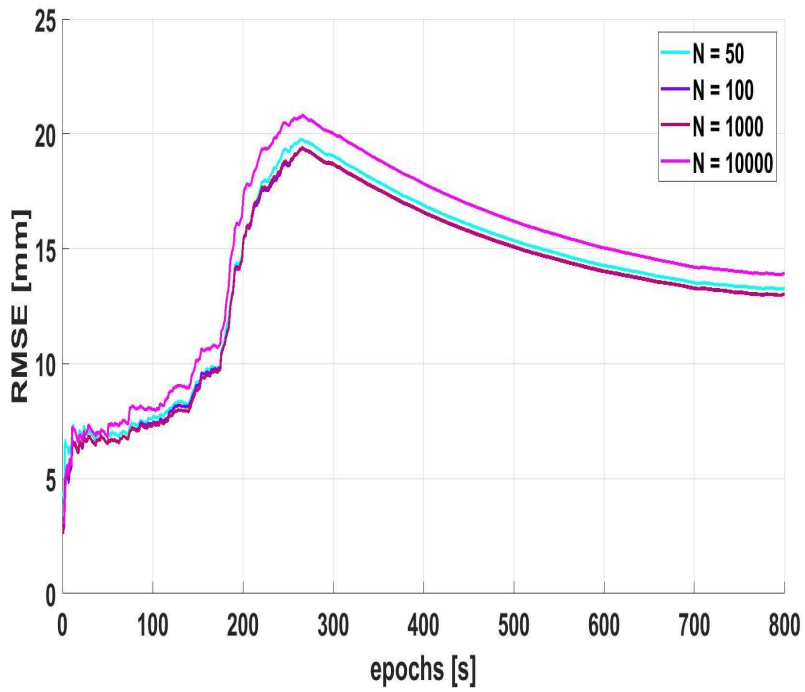

Figure 10

Particle Filter (PF)

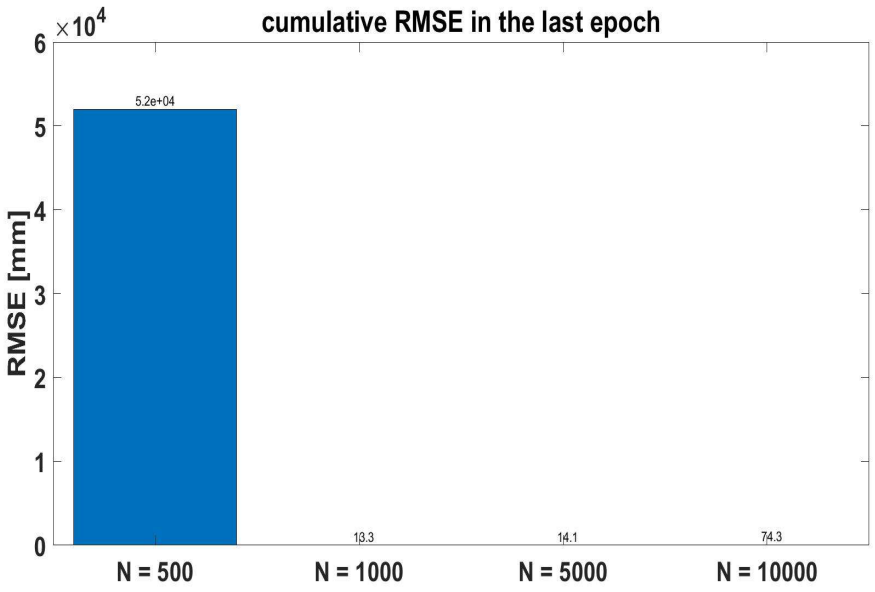

Figure 11

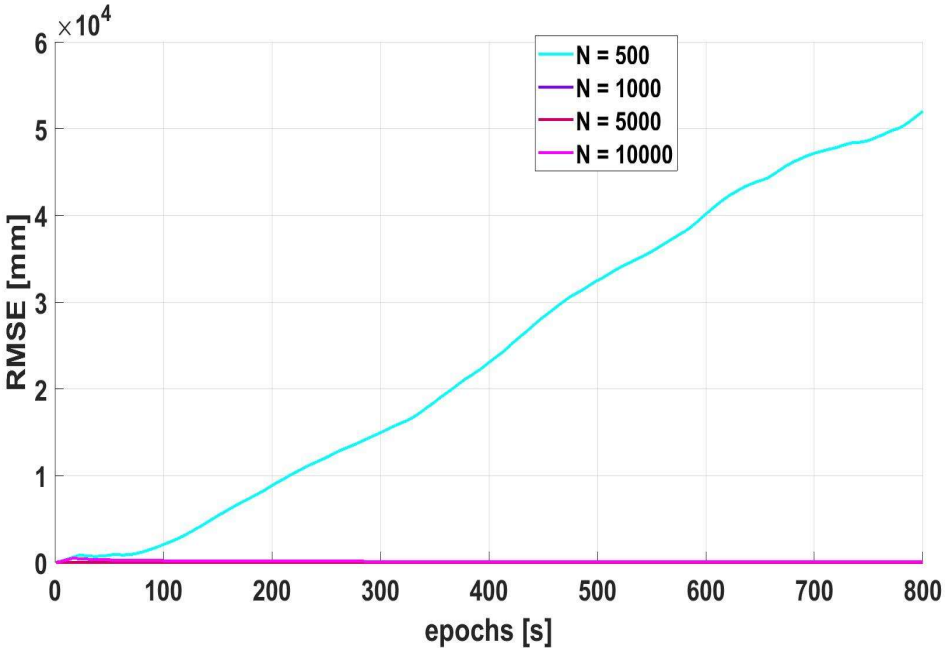

Figure 12